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The General-Equilibrium Effects of the Residential Property Tax: Incidence and Excess Burden

ARTHUR M. SULLIVAN

Graduate School of Administration and Department of Economics, University of California, Davis, California 95616

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An urban general-equilibrium model is used to explore the efficiency effects and distributional consequences of the residential property tax. In a model of a "small," "open" city, the replacement of a nondistortionary land tax with the residential property tax reduces the land area, density, and population of the city. The aggregate net return on land falls by an amount equal to 164.93% of total tax revenue. In a model of a "closed" region, when city Q replaces the land tax with the residential property tax, production and employment shift to other cities in the region. The net return on land decreases in city Q, and increases in other cities in the region; in aggregate, landowners are made better off by the change in tax policy, while residents are made worse off. The excess burden of the property tax is computed as 6.5197% of tax revenue. © 1985 Academic Press, Inc.

1. INTRODUCTION

In this paper, an urban general-equilibrium model is used to explore efficiency effects and distributional consequences of the residential property tax. The paper integrates three strands of tax analysis. The first strand, developed by Grieson [7], Leroy [8], Arnott and Mackinnon [1], and Carlton [4], is concerned with the effects of the property tax on the spatial distribution of residents; ignored in these analyses are the effects of the tax on the urban labor market. The second strand of analysis, developed by Polinsky and Rubinfeld [11] and Brueckner [3], is concerned with the labor-market effects of the residential property tax. The analysis from these models is incomplete: analytical tractability comes at the expense of realism, and effects of the property tax on a number of city characteristics (e.g., population) are indeterminate. The third strand of analysis, developed by Fullerton *et al.* [5, 6, 12–15], is concerned with the computation of equilibria under different tax regimes: using nonspatial models, these authors compute the efficiency and distributional effects of alternative tax regimes.

The analysis of this paper integrates the three strands of tax analysis. An urban spatial-equilibrium model is used to compute equilibria under different tax regimes. The model captures the effects of the residential property tax on the urban labor market (both supply and demand) and the urban land market.

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The results of the analysis can be summarized as follows. In a model of a small open city, the replacement of a nondistortionary land tax with a residential property tax reduces the land area, density, and population of the city. The aggregate net return on land decreases by an amount equal to 164.93% of total tax revenue. In a model of a closed region, when city Q replaces the land tax with the residential property tax, production and employment shift to other cities in the region. The net return on land decreases in city Q, and increases in other cities in the region; in aggregate, landowners are made better off by the change in tax policy, while residents are made worse off. The excess burden of the property tax is computed as 6.5197% of total tax revenue.

2. THE MODEL

In this section, I describe a model of a small, open, monocentric, segregated city. The city is open in the sense that labor and capital are perfectly mobile within and between cities. In equilibrium, the utility level and the price of capital do not vary across space. The city is small in that it is one of many cities. The utility level and the price of capital are determined in national markets, and are exogenous to the city. The city is monocentric, with a central export node through which all export output passes. The city is segregated in the sense that the labor market is separated geographically: labor demand comes from export firms in the central business district (CBD), and labor supply comes from households in a residential fun district (RFD) surrounding the CBD.

The relative positions of all locations in the city will be described by a single variable, u, equal to the radial distance to the CBD. The use of a one-dimensional index of location follows from the assumption that circumferential travel is costless. For the analytical exercises, u is a continuous variable. In the simulation exercises, u is a discrete variable: ring u is an annulus of width w, the midpoint of which is $u \cdot w$ units of distance from the CBD.

The market for land is competitive. The land in a flat, featureless plain is owned by absentee landlords who rent land to the highest bidder. Three sectors compete for land in the plain: the export–production sector, the residential sector, and the agricultural sector.

The CBD is defined as the area over which the export-industry bid-rent P_T^I exceeds all other bid-rents. P_T^I is defined by the condition that profits at all locations equal zero: given transport costs, the price of the export good, and the prices of inputs, $P_T^I(u)$ is the land rent for location u such that profit at location u is zero. P_T^I varies spatially for two reasons. First, transport costs per unit of output rise with distance to the export node, causing P_T^I to decrease as u increases. Second, the price of labor falls as u

increases, since workplaces at large u require shorter commuting trips. As wages fall, they cause P_T^I to increase.

The RFD is defined as the area over which the land bid-rent of the housing-supply sector exceeds all other land bid rents. The housing land bid-rent generates zero profits for housing suppliers, meaning that P_T^H is positively related to housing price. Housing price falls as u increases, since residents at more remote locations have longer commuting trips.

2.1 Residential Sector

The head of every household commutes daily from his residential location in the RFD (u_r) to his workplace in the CBD (u_w). Travel through ring u requires t_T hours and t_H dollars. For computational reasons, the commuting trip is divided into two parts: intra-RFD commuting comes at the expense of leisure time, while intra-CBD commuting time comes at the expense of labor time. Labor is the sole source of income. The utility level of households depends on housing consumption (H), consumption of an import good (X), and leisure time (J_F). Utility maximization generates an indirect utility function of the form:

$$V = V(J_F(u_r), P_H(u_r), P_X, Y(u_r, u_w)); \tag{1}$$

$$J_F(u_r) = \bar{J}_F - t_T \cdot (u_r - u_x); \tag{2}$$

$$Y(u_r, u_w) = P_L(u_w) \cdot J_L(u_w) - t_H \cdot (u_r - u_w); \tag{3}$$

$$J_L(u_w) = \bar{J}_L - t_T \cdot (u_x - u_w); \tag{4}$$

where

$$\bar{J}_F = \text{maximum leisure time};$$

$$u_x = \text{CBD radius};$$

$$P_L(u_w) = \text{wage at } u_w; \text{ and}$$

$$J_L(u_w) = \text{labor time at } u_w.$$

The indirect utility function has as its arguments the price of the import good (P_X), housing price (P_H), leisure time (J_F), and net income (Y). The price of the import good is exogenous and spatially invariant. Housing price is endogenous and varies spatially: the more accessible residential locations command higher housing prices. Leisure time equals the maximum leisure time ($\bar{J}_F$) less intra-RFD commuting time. The time required to pass through ring u is a constant t_T. Net income equals labor income less monetary commuting costs. Labor time per worker at u equals the maximum work time ($\bar{J}_L$) less intra-CBD commuting costs. The wage varies spatially to make workers indifferent among all employment locations: to

the extent that intra-CBD time and money commuting costs are higher, wages are higher; in equilibrium, all workers have the same net labor income (labor income less intra-CBD commuting costs).

In equilibrium, households must be indifferent among all residences and workplaces. That is, utility is constant over u_r and u_w. The equilibrium housing-price function, $P_H(u_r)$ decreases as u_r increases: locations that involve relatively large commuting expenses command lower housing prices. The housing-price gradient, derived by differentiating (1) with respect to u_r (holding utility constant), is

$$\frac{\partial P_H}{\partial u} = \frac{-t_H - q(u) \cdot t_T}{H(u)}, \tag{5}$$

where $q(u)$ equals the opportunity cost of a unit of commuting time. The equilibrium wage function $P_L(u_w)$ decreases as u_w increases, reflecting the fact that workplaces at large u_w involve relatively small commuting expenses. The wage gradient, derived by differentiating (1) with respect to u_w (holding utility constant), is

$$\frac{\partial P_L}{\partial u} = \frac{-t_H - P_L(u) \cdot t_T}{J_L(u)}. \tag{6}$$

2.2 *Housing Production Firms*

Housing is a competitive industry, with profits at all locations equal to zero. Firms produce housing with capital and land under conditions of constant returns to scale. The unit costs function and unit profit function are

$$C_H = C_H\left(P_K, P_T^H(u)\right); \tag{7}$$

$$0 = P_H(u) - C_H\left(P_K, P_T^H(u)\right). \tag{8}$$

P_K equals the cost of capital and P_T^H equals the bid-rent of the housing production sector. The unit profit function implicitly defines the land bid-rent of housing producers as the price of land such that profits are zero. Differentiation of (8), using the properties of the unit cost function, gives the slope of the bid-rent function:

$$\frac{\partial P_T^H}{\partial u} = \frac{\partial P_H / \partial u}{T_H(u)}; \tag{9}$$

where T_H equals land per unit of housing. The slope of the bid-rent function is negative, since the housing price gradient is negative.

2.3 Industrial Sector

There is a single industry (A) in the CBD. The output price of the industry is exogenous. The output of the industry is produced under conditions of internal constant returns to scale and external scale economies. That is, average production costs are constant over the individual firm's output, but decline as industry-wide output increases.

Industry A produces output with inputs of capital, labor, and land. Its production technology is summarized in the unit cost function

$$C_A = C_A(P_K, P_L(u), P_T(u)) \cdot \phi(A^Z), \tag{10}$$

where

$\phi(A^Z)$ = scale-economy variable;

A^Z = aggregate output of industry A; and

$\phi'(A^Z) < 0$.

The unit profit function is

$$\pi_A(u) = 0 = P_A - t_A \cdot u - C_A(P_K, P_L(u), P_T(u)) \cdot \phi(A^Z), \tag{11}$$

where

P_A = exogenous output price; and

t_A = transport costs per unit distance.

The zero-profit condition implicitly defines the land bid-rent of industry A, $P_T^A(u)$, the slope of which is

$$\frac{\partial P_T^A}{\partial u} = \frac{-t_A \cdot \phi^{-1}(A^Z) - L_A(u) \cdot (\partial P_L/\partial u)}{T_A(u)}, \tag{12}$$

where

$L_A(u)$ = labor per unit output of A; and

$T_A(u)$ = land per unit output of A.

Because the wage gradient is negative, the rent gradient is ambiguous in sign. I will assume that the transport costs of A's output are large relative to the cost of transporting labor, i.e.,

$$-t_A < \phi(A^Z) \cdot L_A(u) \cdot (\partial P_L/\partial u), \tag{13}$$

meaning that the rent gradient is negative.

2.4 Government Sector

The city government provides a pure public good, G, the level and cost of which are exogenous. The revenue requirement of the city is therefore fixed at $\bar{R}$ per time period. In the initial equilibrium city, revenue is raised through a nondistortionary land tax. The land tax is based on differential land rent, i.e., the excess of the urban land rent over the exogenous agricultural land rent. The tax on an acre of land at u is

$$\text{tax}(u) = \left(P_{\text{T}}(u) - \bar{R}_{\text{A}}\right) \cdot \tau_{\text{T}}, \tag{14}$$

where: $P_{\text{T}}(u)$ equals the equilibrium urban land rent (either $P_{\text{T}}^{\text{A}}(u)$ or $P_{\text{T}}^{\text{H}}(u)$); $\bar{R}_{\text{A}}$ equals the exogenous agricultural bid-rent; and τ_{T} is the tax rate.

The differential land tax is nondistortionary. Because both the industrial sector and the household sector are taxed at the same rate, the border between the two sectors, u_{x}, is unaffected by the tax. Because the tax is based on differential land rent, the city border, $\bar{u}$, is unaffected by the tax: as P_{T} becomes close to $\bar{R}_{\text{A}}$, the tax approaches zero. Since the tax does not affect the supply of land in urban uses, it is nondistortionary, i.e., it generates no excess burden.

The alternative to the land tax is a residential property tax, i.e., a tax on capital and land used by the housing sector. The property tax is an *ad valorem* tax on land and improvements at rate τ. The tax per unit of housing at location u is

$$\text{tax}(u) = \tau \cdot \left[P_{\text{K}} \cdot K_{\text{H}}^{*}(u) + P_{\text{T}}(u) \cdot T_{\text{H}}^{*}(u)\right] \tag{15}$$

where $I_{\text{H}}^{*}(u)$ is the profit-maximizing quantity of input I. As shown in Section 3, the property tax generates a number of distortions in the urban economy.

2.5 Equilibrium Conditions

General equilibrium is defined as the simultaneous satisfaction of two border conditions and two partial-equilibrium conditions. The two border conditions define the territories of the export industry (the CBD) and the residential sector (the RFD). The CBD border, u_{x}, is the point at which the negatively sloped bid-rent of A equals that of the household sector:

$$P_{\text{T}}^{\text{A}}(u_{\text{x}}) = P_{\text{T}}^{\text{H}}(u_{\text{x}}). \tag{16}$$

The CBD is a circular area of radius u_{x}. The city border, $\bar{u}$, is the point at which the bid-rent of the housing industry equals the exogenous nonurban bid-rent, $\bar{R}_{\text{A}}$:

$$P_{\text{T}}^{\text{H}}(\bar{u}) = \bar{R}_{\text{A}}. \tag{17}$$

The residential district is an annulus of width $(\bar{u} - u_{\text{x}})$.

The first partial-equilibrium condition states that the city's labor market clears. Labor demand comes from the export industry in the CBD. Aggregate labor demand, L^{D}, is

$$L^{\mathrm{D}} = \int_0^{u_x} l_{\mathrm{A}}(u') \cdot g(u')\, du', \tag{18}$$

where $l_{\mathrm{A}}(u)$ equals labor per unit land of industry A at location u; and $g(u)$ equals land area per unit width of ring u. Aggregate labor supply, L^{S}, is

$$L^{\mathrm{S}} = \int_{u_x}^{\bar{u}} \frac{\hat{J}_{\mathrm{L}}(u')}{T_{\mathrm{H}}(u') \cdot H(u')} \cdot g(u')\, du', \tag{19}$$

where

$\hat{J}_{\mathrm{L}}(u)$ = labor supply (in hours) per household;

$T_{\mathrm{H}}(u)$ = land per unit housing in u;

$H(u)$ = housing demand per household in u.

The labor market is in equilibrium if $L^{\mathrm{D}} = L^{\mathrm{S}}$.

The second partial-equilibrium condition concerns the government sector. The city government is assumed to have a fixed revenue requirement of $\bar{R}$. An increase in taxes of one form will cause an equal reduction of other taxes. In the initial-equilibrium city, the sole source of tax revenue is the tax on differential land rent:

$$\bar{R} = \int_0^{u_x} \tau_{\mathrm{T}} \cdot \left(P_{\mathrm{T}}^{\mathrm{A}} - \bar{R}_{\mathrm{A}}\right) \cdot g(u')\, du' + \int_{u_x}^{\bar{u}} \tau_{\mathrm{T}} \cdot \left(P_{\mathrm{T}}^{\mathrm{H}} - \bar{R}_{\mathrm{A}}\right) \cdot g(u')\, du'. \tag{20}$$

In the property-tax city, the land tax is replaced by a tax on capital and land in the housing sector. Total tax revenue (where $k_{\mathrm{H}}^* = K_{\mathrm{H}}^*/T_{\mathrm{H}}^*$) is

$$\bar{R} = \int_{u_x}^{\bar{u}} \tau \cdot \left[P_{\mathrm{K}} \cdot k_{\mathrm{H}}^*(u') + P_{\mathrm{T}}^{\mathrm{H}}(u)\right] \cdot g(u')\, du', \tag{21}$$

where τ = the property tax rate. Since the revenue requirement $\bar{R}$ is exogenous, the property tax rate τ is endogenous: τ is set at a level such that, in equilibrium, the exogenous revenue requirement is satisfied.

2.6 Functional Forms and Parameter Values

In this subsection, I describe the numerical features of the general-equilibrium model. The functional forms and parameter values of the utility function and unit cost functions are described. In addition, the characteristics of the initial-equilibrium city are discussed.

Residential sector. Residents maximize utility, given a Cobb–Douglas utility function whose arguments are leisure time (J_F), housing consumption (H), and import consumption (X). The maximization process generates an indirect utility function of the form

$$V = J_F^{\varepsilon_1} \cdot P_H^{\varepsilon_2} \cdot P_X^{\varepsilon_3} \cdot Y. \tag{22}$$

The demand functions for H and X follow from the differentiation of (22) with respect to income and the commodity price, using Roy's Identity. The income and price elasticities of demand are equal to 1.0 (in absolute value). The budget shares of the two goods are determined by the exponents of P_H and P_X. I assume that 25% of the median household's income is spent on housing. The exponent of J_F determines the monetized value of the marginal disutility of commuting time. I assume that the disutility of commuting time is one-fourth of the wage, making $\varepsilon_1 = 0.25$.

There are two components of commuting costs: monetary costs and time costs. The monetary cost of commuting, t_H, equals \$0.20 per radial mile. The time cost of commuting (t_T) equals 5.0 min per radial mile, meaning that rush-hour traffic moves at a speed of 12.0 radial miles per hour.

Housing is produced, using capital and land, with a CES production function. The CES production function generates a unit cost function of the form

$$C_H(P_K, P_T) = A_H^{-1} \cdot \left[\left(\frac{P_K}{b_K}\right)^{\rho/\rho+1} + \left(\frac{P_T}{b_T}\right)^{\rho/\rho+1}\right]^{\rho+1/\rho}, \tag{23}$$

where P_K is the price of capital and P_T is the price of land. The value of ρ determines the elasticity of substitution between capital and land; ρ equals 0.333, meaning that the elasticity of substitution is 0.75.

The values of A_H, b_K, and b_T determine the budget shares of capital and land. Given the conclusions of Meisekowski [9] and Netzer [10] I assume that, for the median household, capital costs are responsible for 85.64% of housing production costs, and land costs are responsible for 14.36% of housing production costs.

The parameters of the utility function and the unit cost function determine residential density. The price of housing (P_H) and the demand for housing (H) are determined by the parameters of the indirect utility function. The price of residential land (P_T^H) and land demand (T_H) are determined by the parameters of the unit cost function. The parameters of

the two functions were set at levels such that average residential density equals 2359.15 households per square mile (3.68 households per acre): a residential annulus of 3.582 mile width houses 250,038 households.

Industrial sector. The output of the industry A is produced, using capital, labor, and land, with a CES production function. The corresponding unit cost function is of the form

$$C(P_K, P_L, P_T) = A_A^{-1} \cdot \left[\left(\frac{P_K}{a_K} \right)^{\rho/\rho+1} + \left(\frac{P_L}{a_L} \right)^{\rho/\rho+1} + \left(\frac{P_T}{a_T} \right)^{\rho/\rho+1} \right]^{\rho+1/\rho} \cdot \phi(A^Z). \quad (24)$$

The elasticities of substitution between all input pairs is 0.75, and the degree of increasing returns ($\phi' \cdot A^Z/\phi$) is 0.05. The values of A_A, a_K, a_L, and a_T determine the budget shares of capital, labor, and land. Given the conclusions of Meiszkowski [9] and Netzer [10], I assume that, at the median workplace, budget shares are as follows: capital, 25.94%; labor, 69.12%; land, 4.94%.

The parameters of the unit cost function determine the bid-price for land and factor demands. Factor demands, in turn, determine employment density. The values of these parameters are specified in order to generate aggregate labor demand of 250,038 laborers in a CBD of 2.916-mile radius.

The general equilibrium model is solved with a Fortran-77 program on a VAX-11 computer. The computational procedure is described in Sullivan [16].

2.7 Characteristics of the Initial Equilibrium City

Table 1 lists the characteristics of the initial-equilibrium city. The city has a radius of 6.498 miles and a population of 250,038 households. The price of land falls from $91,822 per acre per year at the city center to $1645 per acre at the city's edge. Both the price of labor and the price of housing decrease as u increases. Reflecting the spatial variation in prices, both employment density and household density decrease as u increases. The median household has a one-way radial commute of 3.18 miles, which takes 0.265 hours.

3. THE GENERAL-EQUILIBRIUM EFFECTS OF THE RESIDENTIAL PROPERTY TAX

In this section, I use a model of a small open city to examine the effects of the property tax on the city in which the tax is imposed. Ignored in these two sections are the effects of the tax on other cities. As shown below, the property tax causes emigration from the taxing city; it is assumed that the emigrants simply disappear into an infinitely large "rest of the world." While this emigration will affect the cities to which emigrants migrate, the

TABLE 1

Characteristics of Initial-Equilibrium City

Land allocation	
City radius	6.498 miles
CBD radius	2.916 miles
RFD width	3.582 miles
Land area	132.651 square miles
Population	
Total households	250,038
Average employment density	9,360.127 laborers per square mile
Average residential density	2,359.152 households per square mile

Spatial prices and factor ratios

	City center	Median workplace	CBD border	Median residence	City border
Price of labor	\$10.784	\$10.377	\$10.0	—	—
Price of land (per acre per year)	\$91,822	\$28,999	\$4,658	\$2,886	\$1,645
Price of housing	—	—	21.572	20.060	18.574
Capital : labor	0.2001	0.1946	0.1893	—	—
Capital : land	7.9577	3.3630	0.8551	0.3774	0.2474
Labor : land	40.400	17.2803	4.5183	—	—
Residential density (per square mile)	—	—	3,567.20	2,428.20	1,611.45

Median household	
One-way commuting distance	3.180 radial miles
Weekly labor income	\$402.83
Weekly leisure time	38.53 hr

Input costs as share of production costs

	Median workplace	Median residence
Capital	0.2594	0.8564
Labor	0.6912	—
Land	0.0494	0.1436

analysis of such effects is left for Section 4. For this section, we focus on the effects of the property tax on the taxing city.

The residential property tax is paid, statutorily, by housing suppliers, since all households are assumed to be renters. The unit profit function of the housing-supply sector is

$$\pi(u) = P_{\mathrm{H}}(u) \cdot (1 - \tau) - C_{\mathrm{H}}(P_{\mathrm{K}}, P_{\mathrm{T}}(u)). \tag{25}$$

The zero-profit condition implicitly defines the bid-rent for residential land:

$$\tilde{P}_{\mathrm{T}}^{\mathrm{H}}(u) = \frac{P_{\mathrm{H}}(u) \cdot (1 - \tau) - P_{\mathrm{K}} \cdot K_{\mathrm{H}}^{*}(u)}{T_{\mathrm{H}}^{*}(u)}, \tag{26}$$

where $I_{\mathrm{H}}^{*}(u)$ is the profit-maximizing quantity of input I. The derivative of $\tilde{P}_{\mathrm{T}}^{\mathrm{H}}$ with respect to τ is

$$\frac{\partial \tilde{P}_{\mathrm{T}}^{\mathrm{H}}(u)}{\partial \tau} = \frac{-P_{\mathrm{H}}(u)}{T_{\mathrm{H}}^{*}(u)} < 0. \tag{27}$$

The bid-rent for land is a decreasing function of τ: for fixed P_{H} and P_{K}, an increase in taxes will reduce the amount that the competitive suppliers can pay for land; since land is the residual claimant, it absorbs the tax. The price of housing is fixed because the utility of households is exogenous; in the absence of changes in the values of other arguments of the utility function, P_{H} will not change.

The tax-induced decrease of the housing suppliers' land bid-rent decreases aggregate labor supply for two reasons. First, as $\tilde{P}_{\mathrm{T}}^{\mathrm{H}}(u)$ decreases, the residential sector is outbid for land by the agricultural and industrial sectors: $\bar{u}$ decreases and u_{x} increases; the RFD shrinks, and labor supply is summed over a smaller area. Second, as the price of land within the smaller residential district decreases, housing producers substitute land for capital: $T_{\mathrm{H}}(u)$ increases at all u; residential density decreases as land per unit of housing increases. In (19), the integrand decreases.

The tax-induced shift of the aggregate labor-supply curve causes excess demand for labor. Wages in the city increase, and the economy (a) moves upward along the positively sloped aggregate supply curve; and (b) moves upward along the negatively sloped aggregate demand curve. Equilibrium is restored with higher wages and a net decrease in population.

On the supply side, the wage increase induces in-migration that bids up the price of housing. As P_{H} increases, the residential bid-rent increases. As housing and land prices increase, the residential district increases in size and density, increasing the aggregate supply of labor.

As wages increase, the economy moves along the negatively sloped aggregate demand curve. The increase in wages causes production costs to increase; to maintain zero profit, the bid-rent for industrial land, $\tilde{P}_{\mathrm{T}}^{\mathrm{I}}$ decreases. As $\tilde{P}_{\mathrm{T}}^{\mathrm{I}}(u)$ decreases, labor demand decreases for two reasons. First, the industrial sector is outbid for land by the residential sector: u_x decreases, and the territory of the industrial sector decreases. Second, as the price of labor rises relative to the price of land, labor : land ratios decreases: $l_{\mathrm{A}}(u)$ decreases at all u, and employment density decreases.

Changes in aggregate export output reinforce the changes in aggregate labor demand caused by increased wages. Since the export industry is subject to external scale economies, production costs increase as aggregate output decreases. As production costs increase, the land bid-rent of the industrial sector decreases, causing reductions of the land area and employ-

ment density of the CBD. These scale-economy effects exert downward pressure on aggregate labor demand.

The residential property tax distorts behavior, and therefore generates excess burden, in a number of ways. In the residential district, the tax affects the prices of housing and land, affecting the density and the land area of the residential district, as discussed by Grieson [7], Leroy [8], Arnott and MacKinnon [1], and Carlton [4].

The analysis of this section provides a new perspective on the excess burden of the residential property tax. By considering the effects of the tax on the spatial labor market, additional distortions of the property tax are discovered. First, since the tax reduces city size, the intercity distribution of population is distorted. Second, the tax reduces aggregate export output, and increases production costs as scale economies are lost. Third, the tax distorts the prices of labor and industrial land, reducing the land area and density of the production area of the city. In general, the residential property tax distorts behavior in the CBD as well as in the residential district; the analysis of this paper provides a more complete examination of the excess burden of the property tax.

The analysis of this section also provides a new perspective on the incidence of the residential property tax. The analysis of the labor-market effects of the tax in a small, open city generates the following conclusions with respect to the incidence of the tax. First, although the tax increases the price of housing, households are no worse off, because wages are also higher: housing prices rise in response to the increase in wages. Second, the property tax is borne, in part, by CBD landowners: the tax-induced increase in wages reduces CBD land rent. In addition, CBD landlords incur the efficiency cost associated with the tax-induced decrease in aggregate export output and the resultant loss of scale economies: as production costs increase, the price of land decreases.

General equilibrium computations. In this subsection, I describe the results of computational exercises that demonstrate the general-equilibrium effects of the residential property tax. Table 2 lists the characteristics of cities A and B. In city A, a 3.495% *ad velorum* tax on differential land rents raises $374,555 per week, or $1.498 per household per week. In city B, a 1.693% tax on residential property raises the same total revenue.

The property tax reduces the land area, density, production, and population of the city. In the residential district, the land area decreases while the median lot size increases. In the CBD, the land area and the median labor: land ratio decreases. Total export output decreases by 10.2%, causing average production costs to increase as scale economies are lost.

In the small, open, city, the residential property tax is borne entirely by landowners. Laborers escape any effects of the tax because the supply of

TABLE 2
City Characteristics

	A	B	Q	R/S
Tax revenue ($ per week)	374,555	374,555	374,555	374,555
Population (households)	250,038	225,853	234,275	257,919
Land rent				
CBD ($ per week)	7,053,134	6,558,97	6,532,571	7,314,551
RFD ($ per week)	3,580,168	3,163,577	3,293,023	3,697,145
Agricultural ($ per week)	84,947	303,409	231,600	25,122
Total ($ per week)	*10,718,249*	*9,725,959*	*10,057,193*	*11,036,818*
Median land rent in CBD (per acre per year)	28,999	26,016	27,038	29,328
Median land rent in RFD (per acre per year)	2,886	2,761	2,783	2,896
Spatial prices				
Median wage ($ per hour)	10.377	10.380	10.379	10.366
Base wage ($ per hour)	10.000	10.020	10.013	9.993
Median housing price	20.060	20.271	20.282	20.071
Median factor ratios				
Capital : labor	0.1946	0.1947	0.1947	0.1944
Capital : land (industrial)	3.360	3.106	3.200	3.378
Capital : land (residential)	0.3774	0.3651	0.3674	0.3784
Labor : land	17.280	15.95	16.436	17.372
Lot size (acres)	0.2636	0.2659	0.2640	0.2609
Land area (miles)				
Radius of CBD	2.916	2.850	2.874	2.934
Width of RFD	3.582	3.378	3.444	3.636
Radius of city	6.498	6.228	6.318	6.57

laborers to the city is perfectly elastic at the exogenous utility level. Export firms and their customers escape the tax because firms make zero profits, and sell their output at an exogenous price. Capitalists escape the tax because the supply of capital to the city is perfectly elastic at an exogenous price. In the open-city model, the tax is borne by the owners of the inelastically supplied input, land. For the city as a whole, the residential property tax decreases the net return on land by an amount equal to 164.93% of total tax revenue.

4. THE RESIDENTIAL PROPERTY TAX IN A MODEL OF A CLOSED REGION

In this section, I use a model of a closed region to examine the effects of city Q's property tax on city Q and other cities in the region. The model is designed to capture both the intracity and intercity effects of the property tax. Unlike the analysis in Section 3, the effects of emigration from the

taxing city are analyzed. The emigration of people will increase the demand for land in other cities; the decrease in the net return on land in Q will be offset by increases in the net return on land in other cities. These interurban effects are similar to the capital-market effects discussed by Bradford [2].

The closed region model is distinguished by its mobility assumptions. While intraurban and intraregional migration of households is costless, interregional migration is impossible. That is, households are free to move within and between the region's cities, but cannot leave the region. As a result, in equilibrium, the utility level is the same in all cities, and the common utility level is endogenous. The regional population and the number of cities are exogenous, but the population of each city and the intraurban distribution of activity are endogenous.

For simplicity, I will assume that there are three cities in the region. Before the imposition of the property tax in city Q, the cities are identical: they have equal populations, labor productivities, and wages; the spatial distributions of employment and residence are identical. All cities finance the public good with a nondistortionary land tax.

In the three-city closed-region model, emigrants from the property-tax city relocate in cities R and S. The migration to R and S generates excess demand for housing and excess supply of labor. As a result, housing prices rise and wages fall; both price changes reduce the utility level attainable in cities R and S. As the utility level of R and S decreases relative to that of Q, reverse migration occurs, the effect of which is to decrease the utility level in Q as housing pices rise and wages fall. Interurban migration occurs until the utility levels of the three cities are equal.

Regional equilibrium is restored with the following changes in the three cities, listed in Table 2. First, the common (regional) utility level decreases. Second, population shifts from city Q to cities R and S. Third, reflecting the changes in population (changes in the demand for land) the gross (and net) return on land decreases in the property-tax city, and increases in the land-tax cities. Fourth, housing prices increase in all three cities. Fifth, wages increase in Q, and decrease in R and S. In general, intercity migration equalizes the utility levels of the three cities, transmitting the distortionary effects of the property tax to the land-tax cities.

Incidence and excess burden. The welfare loss experienced by households is measured as a general-equilibrium compensating variation payment. Households in the region are made worse off by increases in housing prices and decreases in wages. The amount required to restore the original utility level (the level attained when all cities employ the land tax) is $0.50 per household per week. The welfare loss experienced by the household sector equals $375,057 per week, or 100.13% of the property tax revenue generated in city Q.

In aggregate, landowners experience welfare gains as a result of the change in city Q's tax policy. Although the switch from the land tax to the residential property tax decreases regionwide land rent, it also reduces land taxes. As a result, the net return on land increases by $350,637. Landowners are made better off only because of the substitution of the property tax for the land tax in Q. If, instead, the property tax were introduced into a taxless economy, and its revenue were to be refunded in a nondistortionary manner, landowners would be made worse off as a result of the excess burden of the residential property tax.

Summing the welfare gains and losses of city Q's property tax, the tax generates net welfare losses in the regional economy. In aggregate, the change in tax regimes generates welfare losses of $24,420 per week. Since the change in tax policy leaves total tax revenue unchanged, the dollar figure is the excess burden of city Q's property tax. The excess burden is 6.5197% of the revenue generated by Q's residential property tax.

5. SUMMARY AND CONCLUSIONS

An urban general-equilibrium model is used to explore the effects of the residential property tax. The analysis differs from earlier research efforts in its consideration of the labor-market effects of the property tax in spatial general-equilibrium models. The property tax is analyzed in two models: in the model of a small, open city, attention is focused on the effects of the tax on the taxing city; in the model of a closed region, the effects of tax-induced intercity migration are explored.

The replacement of a nondistortionary land tax with the property tax causes a number of changes in the small, open city. The property tax reduces aggregate labor supply, causing the city's wages and population to decrease. Aggregate export output decreases, providing upward pressure on production costs as scale economies are lost. In response to changes in factor and output prices, employment density and residential density decrease. Landowners (in both the CBD and RFD) are made worse off by the change in tax policy: in the open-city model, they bear the entire burden of both taxes; since the property tax generates excess burden, landowners would prefer the land tax. The property tax reduces the net return on land by an amount equal to 164.93% of total tax revenue.

The replacement of city Q's land tax with a residential property tax causes a number of changes in the closed region. Cities R and S (which employ a nondistortionary land tax) grow at the expense of city Q. As production shifts to R and S, production costs in R and S decrease as scale economies are realized, while production costs in Q increase as scale economies are lost. Wages increase in Q, and decrease elsewhere in the region. Housing prices increase everywhere, with the largest increases in Q. The welfare loss of regional residents is $0.50 per household per week,

totalling $375,057, or 100.13% of Q's property-tax revenue. The net returns of landowners in city Q decrease by 2.189%, while the net returns of landowners in cities R and S increase by 2.99%. The aggregate (region-wide) net return on land increases by 1.117%, or $350,637. In aggregate, the property tax generates an excess burden equal to 6.5197% of city Q's property-tax revenue.

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